

Sustainability Lab

IIT Gandhinagar

Scenario-1 : Earth Observation

The Ministry of Environment has commissioned an AI-based audit of the Delhi Airshed to identify land use patterns and pollution sources. You are provided the following datasets:

- Shapefile of the Delhi-NCR region (EPSG:4326) [\[File\]](#)
- Sentinel-2 RGB image patches (128×128 pixels, 10m/pixel resolution), with each image filename mapped to its center coordinates (latitude, longitude) in **.png** format. [\[File\]](#)
- Shapefile of the Delhi-Airshed region (EPSG:4326) [\[File\]](#)
- A land_cover.tif raster (ESA WorldCover 2021, 10m resolution) [\[File\]](#)

Your goal is to build a full pipeline involving spatial gridding, land cover classification.

Q1. Spatial Reasoning & Data Filtering

- Plot the Delhi-NCR shapefile using matplotlib and overlay a 60×60 km uniform grid
- Overlay this grid on a satellite basemap using geemap or leafmap
- Mark the four corners and the center of each grid cell
- Filter images based on whether their center coordinates fall within the grid
- Count and report the number of images before and after filtering

Q2. Label Construction & Dataset Preparation

- For each image, extract a 128×128 patch from the land_cover.tif centered at the image's coordinate
- Assign a label using the mode (most frequent) land cover class in the patch
- Map ESA class codes to 11 standardized labels
- Handle edge cases and discuss treatment of no-data pixels or mixed class dominance

- Perform a 60/40 train-test split randomly
- Visualize class distribution and discuss balance (or imbalance)

Q3. Model Training & Supervised Evaluation

- Train a CNN classifier (e.g., ResNet18) on the training set
- Evaluate using a custom F1 score implementation
- Evaluate using **torchmetrics.F1Score** and compare results
- Show and explain a confusion matrix
- Plot 5 correct and 5 incorrect predictions with images and labels

Supporting Information

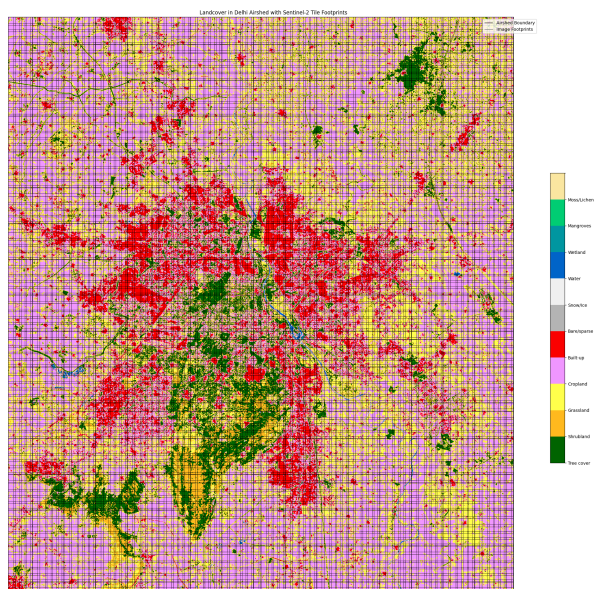
- Image Size: 128×128 pixels
- Resolution: 10m/pixel (Sentinel-2 RGB)
- CRS for Gridding: EPSG:32644
- Label Source: ESA WorldCover 2021 (**land_cover.tif**, 10m resolution)
- VLM Candidates: BLIP (Salesforce), OFA (Microsoft), GIT (Google)
- Visualization SDK: Use **geemap.Map().add_basemap("SATELLITE")**

Label Assignment Logic: Each image's label is derived from the mode value in the corresponding 128×128 label patch extracted from **land_cover.tif**, centered on its coordinate. For instance:

- 8000 pixels = class 50 (Built-up)
- 3000 pixels = class 40 (Cropland)
- 512 pixels = class 10 (Tree cover)

The assigned label would be Built-up (50), as it is the dominant class.

Helping Figures:



ESA WorldCover 2021 - Delhi NCR

